ALEXANDRE PROENÇA

Senior / Staff Software Engineer — Python, Distributed Systems & AI Agents

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SUMMARY

Senior / Staff Software Engineer with 14+ years of experience designing and building distributed backend systems in Python. Currently focused on applying AI design patterns and building production-grade AI agents using LangChain, Retrieval-Augmented Generation (RAG), and vector databases. Strong track record across architecture, technical leadership, and mentorship — from founding and leading an RPA department to guiding teams through code review and design standards. Based in Florianópolis, Brazil.

TECHNICAL SKILLS

AI & Agentic Systems: Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), Agentic Automation, LangChain / LangSmith, AI Design Patterns

Languages: Python, Go (Golang), Elixir, JavaScript

Frameworks: FastAPI, Django (+ DRF), Flask, Bottle, SQLAlchemy

Data Stores: PostgreSQL, MongoDB, MySQL, Redis, Qdrant (Vector DB), Elasticsearch

Messaging & Architecture: Apache Kafka, RabbitMQ, Microservices (RPC, gRPC, HTTP, WebSockets), Service Mesh, Distributed Systems Design

Infrastructure & Observability: Docker / Docker Swarm, Ansible, AWS (EC2, Fargate, S3, SQS, SNS, RDS, Route53, SES, Elastic Beanstalk), Prometheus, Grafana, OpenTelemetry, Graylog

Practices: Technical Mentorship, Code Review, Agile (Scrum/Kanban), BDD, TDD/Pytest

PROFESSIONAL EXPERIENCE

Senior Software Engineer
Jan 2025 – Present

Enterprise Engineering Ltda • *Remote*

- Provide expertise in architectural decisions, code reviews, technical standards, and best practices.
- Apply AI design patterns and build production AI agents.
- Design distributed systems architecture across services.
- Act as reviewer on GitLab repositories and provide technical mentorship.

Technologies: *Qdrant, PostgreSQL, MongoDB, Microservices (RPC/gRPC/HTTP/Sockets), Redis (cache/queue/pub-sub), Apache Kafka, Python, LangChain/LangSmith, FastAPI, SQLAlchemy, Docker, Prometheus, Grafana, OpenTelemetry*

Software Staff Engineer
Jun 2023 – Sep 2024

Solfácil • *São Paulo, Brazil*

- Supported the Solar Loan Tribe, collaborating across three teams on a distributed system for selling solar-energy loans.
- Provided expertise in architectural decisions, code reviews, technical standards, and best practices across teams.
- Acted as reviewer on GitLab repositories and provided technical mentorship.

Technologies: *PostgreSQL, MongoDB, Microservices (RPC/gRPC/HTTP/Sockets), Redis, Apache Kafka, Python, Elixir, FastAPI, SQLAlchemy, Docker, Prometheus, Grafana, OpenTelemetry*

Tech Lead — RPA Department Lead
Jan 2022 – Jun 2023

TEx • *On-premise (UOL infrastructure)*

- Established the RPA department from scratch, leading a team of four.
- Designed the full architecture and implemented a microservices-based solution to automate customer information verification, download documents from insurance companies, and feed data into a SaaS platform.
- Reduced manual effort and errors while saving significant time and cost.

Technologies: *Python, RabbitMQ, MongoDB, FastAPI, Docker Swarm, Ansible, Redis, Graylog*

Senior Software Engineer
Jun 2021 – Jan 2022

TEx • *On-premise (UOL infrastructure)*

- Designed and maintained a distributed system using a service-mesh architecture.
- Coded and reviewed on GitLab repositories across microservices (RPC/gRPC/HTTP/Sockets).

Technologies: *MySQL, MongoDB, Redis, Docker Swarm, Graylog*

Backend Engineer *Jul 2018 – Jun 2021*

OnSign TV — Digital Signage • *Florianópolis, Brazil*

- Developed new features and maintained the SaaS platform using Django (Python).
- Maintained and upgraded frontend applications (JavaScript, HTML5, CSS).
- Maintained and developed the PostgreSQL database layer and Golang microservices.
- Handled transactional integrity and atomic operations in asynchronous event calls.

Technologies: *PostgreSQL, Django, Golang, Redis, Elasticsearch, Ansible*

EARLIER EXPERIENCE

Mentor — Codenation *May 2019 – Nov 2019*

Florianópolis, Brazil

Mentored participants of AceleraDev, Codenation's acceleration program for developers and data scientists, providing feedback and support on programming topics.

Software Developer — Mercado Livre *May 2017 – Jun 2018*

Florianópolis, Brazil

Built and optimized the Fee Calculator API, computing Brazilian shipping fees by postal code and carrier, sustaining traffic spikes of up to 2,000 requests/second. Stack: Django/DRF, Flask, MySQL, Docker, Ansible, Jenkins, Pytest.

Python Developer — SocialBase *Nov 2015 – Jan 2017*

Built a real-time WebSocket messaging API (Tornado) and distributed backend services. Stack: MongoDB, Bottle, RabbitMQ, Locust load testing, BDD, async Python.

Backend Developer — Borayu Tecnologia *Jun 2014 – Nov 2015*

Built web APIs and mobile back-ends on AWS (EC2, SNS, SQS, S3, RDS, Route53, SES, Elastic Beanstalk) with Django/DRF and MySQL; covered by unit, integration, and deployment tests (Selenium, Karma, Jasmine, PhantomJS).

Python Developer (Systems Analyst) — Mandriva *Oct 2011 – Jun 2014*

Curitiba, Brazil

Developed modules for the Mandriva Business Server and the Mandriva Management Console (MMC), a web interface for managing LDAP directories, services, and network resources. Owned the Squid and Apache service-management modules end to end.

EDUCATION

MBA, Software Architecture & Solutions *Mar 2024 – Sep 2025*

Full Cycle

Systems Analysis and Development *2010 – 2013*

Universidade Positivo

Technical Degree, Electronics *1997 – 2003*

Universidade Tecnológica Federal do Paraná (UTFPR)

CERTIFICATIONS

- Redis University — RU101: Redis Data Structures (Redis Labs)
- Building an API with Python and FastAPI
- Introduction to AWS Fargate
- Introduction to EC2 Auto Scaling

LANGUAGES

English — Full Professional Proficiency | Portuguese — Native / Bilingual